

Unit 3: Cells

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3.1. What is a cell?	Define a cell
3.2. Cell theory	Outline the contributions of Theodor Schwann and Matthias Schleiden to the development of the cell theory
3.3. Cell structure and function	Compare and contrast the structures of plant and animal cells
3.4. Types of cells	Differentiate between eukaryotic and prokaryotic cells
3.5. Animal and plant cells	Make temporary slides of plant and animal cells
3.6. Observing cells under a microscope	Observe and draw animal and plant cell parts under the microscope
3.7. The cell and its environment	Conduct substances across animal and plant cells
3.7.1. Passive transport	experiments about the movement of substances across animal and plant cells (osmosis, diffusion)
3.7.2. Active transport	Construct with examples the different levels of biological organization
3.8. Levels of Biological Organization	

3.1. What is a cell?

Recall

Recall grade seven general science lesson:

Unit 4: cell as the basis of life

Objectives

At the end of this section, the student will be able to:

- explain how living things are composed of cells*
- identify unicellular organisms from multicellular organism.*

Activity 3.1: Group work

- Why do we study about cell?
- List some important biological activities take place in the cell
- Discuss in groups about the unicellular and multicellular organisms

Cell is the smallest and the basic unit of living things. All living things are composed of cells. Some of these living things consist of a single cell and are called **unicellular organisms**.

For example, paramecium is a unicellular organism. Most of the unicellular organisms are invisible with naked eyes, but there are some exceptions like chicken egg cells which can be seen with naked eyes. The other groups of living things such as plants and animals including humans which consist of many cells are called **multicellular organisms**. The body of multicellular organisms is composed of many cells of specialized types which work cooperatively. For example, in

the case of human body, everything you do, every action and every thought are possible because of the processes that occur at the cellular level. Each activity of our body is a result of the cooperative work of muscle cells for movement, nerve cells for gathering information and brain cell for interpretation of message to action.

Attention

Most cells are very small and are, therefore, visible only with a microscope but an ostrich egg is a single cell with about 6 inches in diameter and weighing about 1.36 kg; nerve cells in your body can stretch several feet long, and nerve cells in giant squid can be more than 30 feet long!

Key Terms

Cell: is the smallest and the basic unit of living things. All living things are composed of cells. Unicellular organisms: living things consist of a single cell

Multicellular organisms: groups of living things composed of many cells

3.2. Cell theory

Objectives

At the end of this section, the student will be able to:

- *describe cell theory,*
- *name scientists who played a role in the discovery of the cell theory.*

Activity 3.2 : Group work

Discuss in group about the cell theory

An English scientist, Robert Hooke was the first to observe cells and to use the word cell for structures in living organisms. He did this in 1665 after examining thin slices of plant material cork. He was surprised by the regular appearance of the structure, which is 'pore like' regular structures that Hooke called 'cells'. Each cell appeared to be an empty box surrounded by a wall. Hooke had discovered and described a cell, without realizing that it is the fundamental unit of all living things.

After Hooke discovered cells in the first half of the seventh century, other early Dutch Microbiologist Anton van Leeuwenhoek (1632—1723) used his lenses in building numerous microscopes, some with magnifications approaching 300X. In 1674 he likely observed protozoa for the first time and several years' later he observed bacteria. Those "very little animalcules" he was able to isolate from different sources, such as rainwater, pond and well water, and the human mouth and intestine. Leeuwenhoek is universally acknowledged as the father of microbiology because, he discovered micro-organisms like protists and bacteria.

More than 150 years later, German scientists, Matthias Schleiden and Theodor Schwann were the first to explain the basic views of what we now call cell theory.

In 1838, German botanist Matthias Schleiden, a botanist stated that



Figure 3.1. Drawing of cork cells published by Hooke

Attention

English natural philosopher Robert Hooke was the first to observe cells in 1665, naming the shapes he saw in cork cellulae (Latin, "small rooms"). They are known to us as cells.

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all plants are made up of cells, and a year in 1839, another German zoologist Theodor Schwann reported that all animals are made up of cells. The two German biologists, established cells as the natural unit and function in living things, the entire animals and plants are made up of these cells. Then a German pathologist, Rudolf Virchow (1856) established the idea that cells arise only by the division of existing cells. Based on the above discoveries, the principle of cell theory is developed. The cell theory states that:

1. Living organisms are composed of one or more cells.
2. Cells are the smallest unit of life
3. Cells arise only by the division of a pre-existing cell.

Cells vary considerably in size and shape but they share certain common features:

- Every living cell is surrounded by a membrane, which separates the cell contents from everything else outside.
- Cells contain genetic material which stores all of the instructions needed for the cell's activities. Many of these activities are chemical reactions, catalysed by enzymes produced inside the cell.
- Cells have their own energy release system that powers all of the cell's activities, So cells can be thought of as the smallest living structures – nothing smaller can survive.

3.3. Cell structure and function

Objectives

At the end of this section, the student will be able to:

- *identify the main structure of a cell,*
- *state that the cytoplasm of all cells contains most organelles like the nucleus, ribosomes, and rough endoplasmic reticulum,*
- *describe the function of the structure of a cell, and*
- *list the main structure of a cell.*

Cells have different structures which have different functions. The organization of cells varies between different organisms. In multicellular organisms, some cells have the same shape and same function but other cells have different shapes and different functions. For example, in the human body, the shape and function of nerve cells are different from that of muscle cells.

Activity 3.3: Self-test

- 1. Read books and/or search the Internet for the function of the structure of a cell. Present your report to your classmates.
- 2. Complete the following table based on your search

Function	Cell Structure
Control all activity of the cell	
Control the transport of material into and out of a cell	
The place where protein is synthesized	
Packaging of the materials in a cell	
Produce energy	

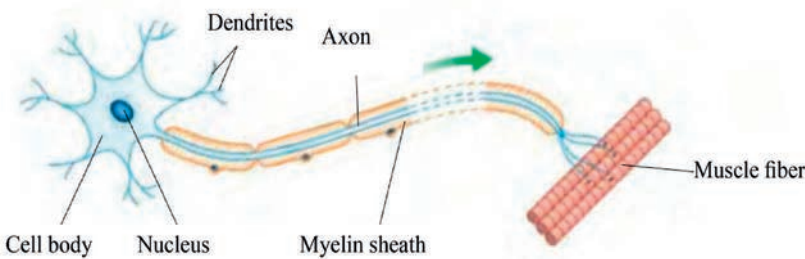


Figure 3.2 Nerve cell

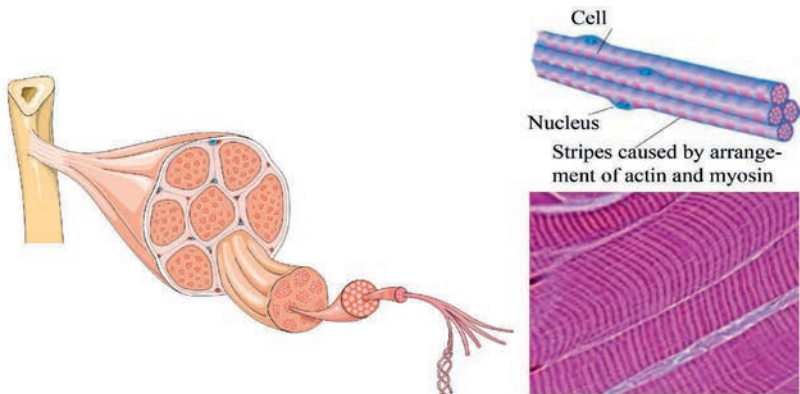


Figure 3.3 Muscle cells

Activity 3.4: Group work

Make a group of five students, examine Fig 3.2 and 3.3 carefully and discuss the structures and functions of the cell.

Cells look like one another in certain important ways. One important similarity between plant and animal cells is the presence of common cell structures. The major structures these cells have in common are:

- 1. Nucleus
- 2. Cytoplasm, and
- 3. Cell (plasma) membrane.

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In addition to the above structures, there are many tiny structures in the cytoplasm called organelles. An organelle is a distinct structure within a cell having a specific function. Some of these organelles have a membrane, while others without membrane.

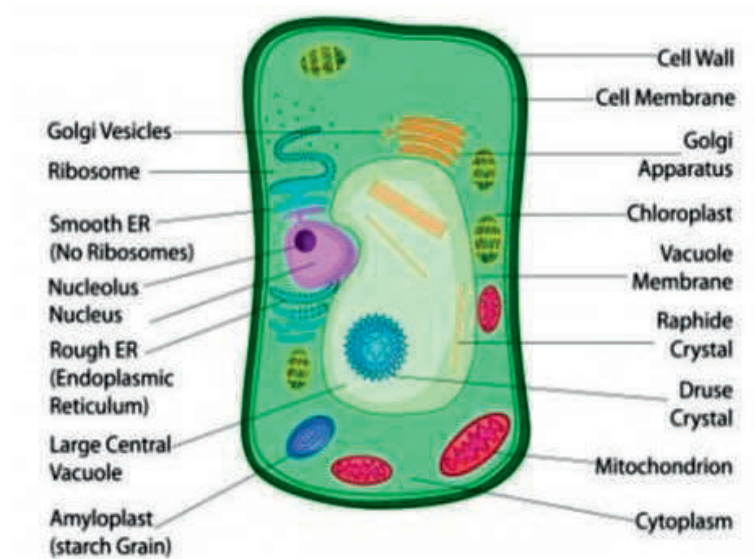


Fig. 3.4 Structure of typical plant cell

Activity 3.5: Think-pair-share

What is the difference between the cell wall and cell membrane? Answer the question in relation to their function, discuss in pair and share your answer with the other classmate.

Cell wall

Cell wall is a rigid structure of a plant cell that provides structural support and gives shape to the cell. It is found external to the cell membrane. Fungal cells, some Protista cells such as algae and bacterial cells also have cell walls. Animal cells have no cell wall.

Cell membrane

Cell membrane is an outer covering that separates the cell's interior from its surrounding environment. Cell membrane is a very thin selectively permeable structure. Cell membrane controls what enters into and leaves a cell. It controls the entry of organic molecules, ions, water, and oxygen into and out of a cell. Wastes, such as carbon dioxide also leave a cell by passing through a cell membrane.

Attention

Cell membrane is semi-permeable (selectively permeable) because it chooses what can go in and out of the cell

Cytoplasm

Cytoplasm is a thick fluid-filled region in the cell containing cell organelles. The fluid material of the cytoplasm consists of 70 to 80 percent water with different organic (proteins, carbohydrates) and inorganic (ions such as sodium, potassium, calcium) molecules

dissolved in it. It is also the site where several chemical reactions such as protein synthesis take place.

Nucleus

Nucleus is the largest organelle surrounded by the double-membrane. Inside a nucleus, there are thread-like structures called chromosomes that contain a very long molecule of DNA (Deoxyribonucleic acid). The nucleus determines what the cell will be, for example, a blood cell, a liver cell, a muscle cell or a nerve cell. The nucleus also controls cell division. Within a nucleus there is a darker area called the nucleolus, this is the site where new ribosomes are made.

Ribosomes

Ribosomes are small cellular structures that are found in large numbers in all cells. The function of ribosomes is to synthesize proteins. They are found either freely in the cytoplasm or attached to the rough endoplasmic reticulum.

Endoplasmic Reticulum

The endoplasmic reticulum is a complex system of interconnected double membranes. They contain fluid-filled spaces between the membranes which allow materials to be transported throughout the cell. There are two distinct types of the endoplasmic reticulum

Rough Endoplasmic Reticulum

It gets its name from the dotted appearance of its surface. These dotted structures on it are the ribosomes. Proteins can be made by these ribosomes. Proteins can be made and stored in the rough endoplasmic reticulum and transported around the cell in the small sacs called vesicles.

Smooth endoplasmic reticulum

This is endoplasmic reticulum without ribosomes. It is important in the manufacturing of lipids.

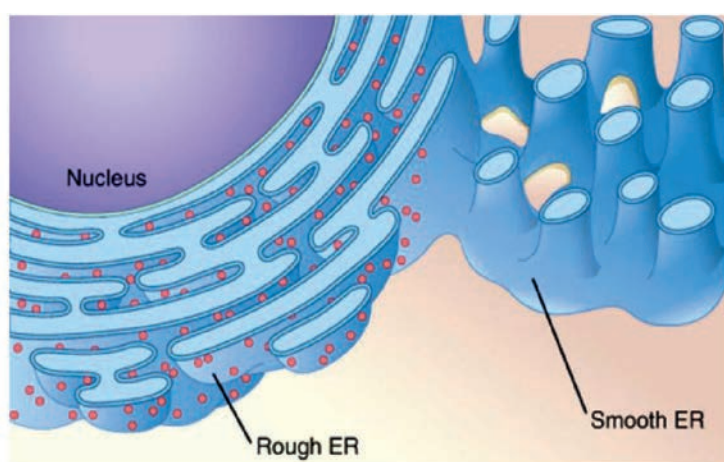


Figure 3.5. The Rough and Smooth Endoplasmic Reticulum

Activity 3.6: Homework / Individual assignment

Using the Internet or books as a source of information, study the relationship between chromosomes, chromatin, and DNA.

Attention

Most cells contain one nucleus but there are interesting exceptions. For example, both the mature red blood cells of mammals and the sieve tube elements of the phloem of flowering plants are without a nucleus. Both lose their nucleus as they mature. The individual cylindrical fibers of voluntary muscle consist of a multinucleate sack. Fungal mycelia also contain multinucleate cytoplasm.

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Activity 3.7: Group work

What makes rough endoplasmic reticulum rough?

Attention

Golgi apparatus is an organelle named by its discoverer (Italian scientist Camillo Golgi), who first described this structure in 1898.

You can think of the Golgi apparatus as a detailing facility that receives ship newly manufactured cars (proteins), puts on the finishing touches, stores the completed cars, and then ships them out when needed. Products made in the ER reach the Golgi apparatus in transport vesicles.

Vesicles

Vesicles are a small sack-like collection of the membranous sac. They store and transport and ship materials where they need to go. For example, a protein made by the rough endoplasmic reticulum may be placed inside a transport vesicle and shipped to the Golgi bodies, where it would be modified and tagged for shipment elsewhere. These vesicles are found in the cell by fusing with the Golgi apparatus.

Golgi apparatus (Golgi body, Golgi complex)

Golgi apparatus is a smooth, concave, membranous structure. The Golgi apparatus with the vesicles functions in the collection, packaging, and distribution of molecules within the cell or even outside the cell.

Proteins and lipids manufactured on the rough and smooth endoplasmic reticulum are transported into the Golgi apparatus and modified as they pass through it. These modified proteins and lipids are then packaged into vesicles that pinch off from the Golgi apparatus. These vesicles then diffuse to other locations in the cell, distributing the newly synthesized molecules to their appropriate place.

Mitochondria

Mitochondria are relatively large organelles found in the cells. It has a double membrane; the outer one controls the entry and exit of materials. The inner membrane forms many folds on which some of the chemical reactions of respiration take place and the site for the synthesis of much energy. Mitochondria are often described as the 'powerhouse' of the cell since they are the sites of respiration.

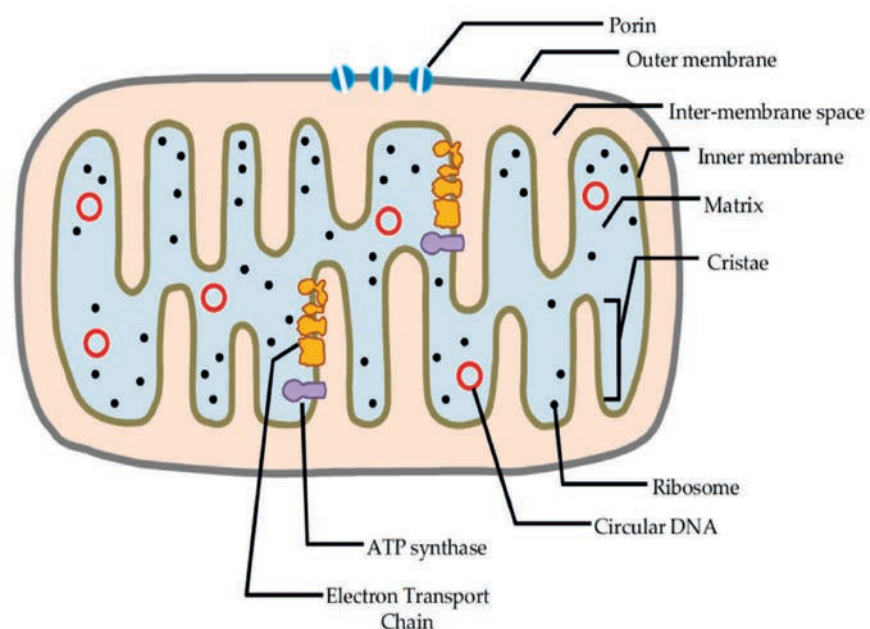


Figure 3.6: Mitochondria

Activity 3.8: Group work

Why do you think some metabolically active cells such as muscle cells contain thousands of mitochondria? Discuss in groups of three to five students and present what you have discussed to your classmates.

Lysosomes

Lysosomes are little membrane-bound packages of digestive enzymes. They contain digestive enzymes which can be used to digest bacteria or other cells taken into the cell. It is also break down unwanted or damaged organelles within the cell.

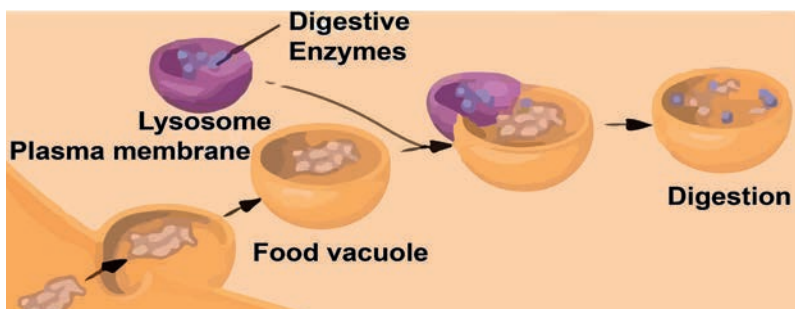


Figure 3.7: Lysosome

Vacuoles

These are organelles that consist simply of a single membrane with fluid inside. Many plant cells have large vacuoles that occupy more than half of the cell volume. Vacuoles are small in the case of animal cell. Vacuoles perform a variety of functions in different kinds of cells. Food vacuole found in some animals to digest food and in the case of some unicellular organisms there is a contractile vacuole which is function is to expel excess water. For example, amoebas and many other unicellular eukaryotes have food vacuole in order to eat smaller organisms or food particles, by a process called phagocytosis. Many unicellular protists living in fresh water have contractile vacuoles that pump excess water out of the cell, thereby keeping a suitable concentration of ions and molecules inside the cell.

Chloroplasts

Chloroplasts are also double membrane-bounded organelles that occur in the cell of green plants, mostly in the cells of the leaves. They are the site for photosynthesis. They contain a green pigment called chlorophyll that can trap the sunlight energy for the synthesis of carbohydrates.

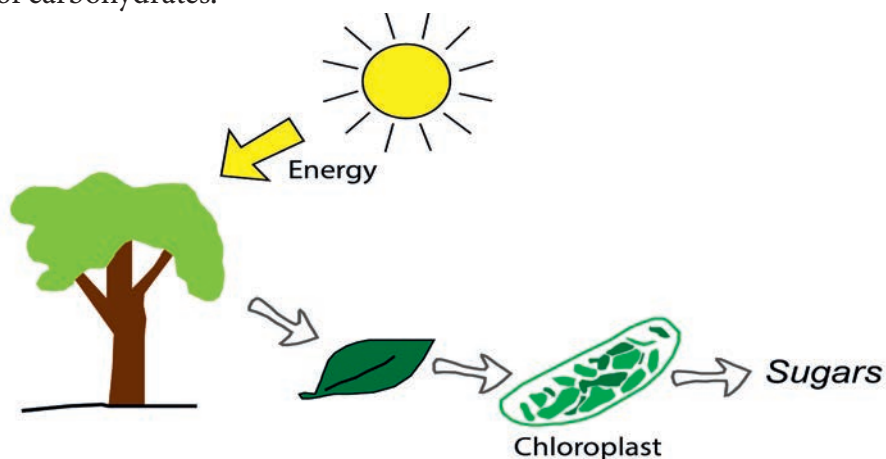


Figure 3.8: Chloroplast trapping sunlight energy to synthesize sugar

Attention

Lysosomes are involved in the breakdown of the contents of 'food' vacuoles. For example, harmful bacteria. Any foreign matter or food particles are taken up into these vacuoles is then broken down by the digestive enzymes. This occurs when lysosomes fuse with the vacuole.

Lysosomes also destroy damaged organelles in this way.

Phagocytosis came from two Greek words phagein which means to eat, and kytos is vessel referring here to the cell.

Activity 3.9: Individual assignment

Make Internet search or refer books and list some unicellular eukaryotes which have food vacuole and contractile vacuole.

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Centrioles and microtubules

The cytoplasm of cells consists of small cylindrical fibres called microtubules that have a variety of roles, including moving chromosomes during cell division. In addition, animal cells have structures called centrioles that anchor to micro-tubules during cell division.

Cilia and flagella

Cilia and flagella are the most common organelles for locomotion in unicellular organisms.

Cilia are short, hair-like structures. Cilia are used for locomotion. It can also be used to create a current in the fluid next to the cell. For example, cilia occur in large numbers on the lining (epithelium) of the air tubes serving the lungs (bronchi).

Flagella are long, thread-like structures, present in lesser number. Flagellum extends from the plasma membrane and enables an entire cell to move. When present, the cell has just one flagellum or a few flagella. When present, cilia are smaller and many.

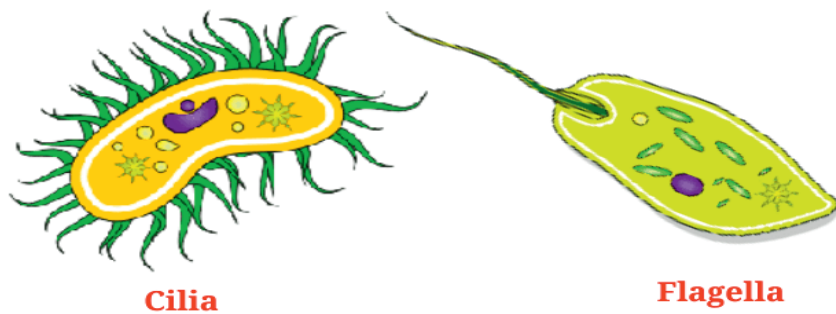


Figure 3.9. Cilia and Flagella

Key Terms

Organelles: small structures found in plant and animal cells. Some examples of organelles are: nucleus, mitochondria, vesicles, and Endoplasmic reticulum.

Cytoplasm: is a thick fluid-filled region in the cell containing cell organelles. It is also the site where several chemical reactions such as protein synthesis take place.

Nucleus: is the largest organelle which contains DNA and control all activity of the cell.

Ribosomes: are small cellular structures that are found in large numbers in all cells. The function of ribosomes is to synthesize proteins

Rough endoplasmic reticulum: contains the protein synthesizing machinery called ribosome.

Vesicles: are small stack-like collections of the membranous sac. They store and transport and ship materials in the cell.

Mitochondria: is the sites of respiration to produce energy.

Lysosomes: contain digestive enzymes which can be used to digest bacteria or break down unwanted substance in the cell.

Chloroplast: are the site for photosynthesis, which contain a green pigment called chlorophyll that can trap the sunlight energy.

3.4 Types of cells

Objectives

At the end of this section, the student will be able to:

- *describe the difference between prokaryotes and eukaryotic cells*
- *give examples of the two types of cell.*

Based on cell structure and cellular organization, cells are grouped into two types. These are Prokaryotic cells and Eukaryotic cells. Prokaryotic cells are usually much smaller than eukaryotic cells. They have a much simpler structure and are thought to be the first cells to have evolved. Prokaryotic cells such as bacterial cells don't have membrane-bound organelles or a nucleus. Eukaryotes such as plants and animals cells, in contrast, have membrane-bound organelles and a nucleus.

Prokaryotic cells

Prokaryotes are different from eukaryotic cells because of the absence of a nucleus and other membrane-bounded organelles. These Prokaryotic cells are generally much smaller than eukaryotic cells. Prokaryotic organisms are unicellular. The cell wall acts as an extra layer of protection, helps the cell maintain its shape, and prevents dehydration.

Eukaryotic cells

Unlike prokaryotes, eukaryotes have a nucleus and other membrane-bound organelles. For example, each organ in an animal's body is specialized to perform a particular role, each organelle in a eukaryotic cell has a distinctive structure and function. Prokaryotic organisms are unicellular but eukaryotic organisms are usually multicellular.

Activity 3.10: Group work

Make a group of three to five students and discuss the structures commonly found in Prokaryotic and Eukaryotic cell, the common cell structures found in both prokaryotic cells and eukaryotic.

Give examples organisms which have prokaryotic cells and eukaryotic

Attention

The term Eukaryotic is derived from the Greek eu, true, and karyon, kernel, referring to the nucleus. Thus Eukaryotic cells are cells with a "true nucleus".

Similarly, the term Prokaryotic is derived from the Greek pro means before and karyon means nucleus; "before nucleus", reflecting the earlier evolution of prokaryotic cells. Eukaryotic cells are usually between 10 and 100 micro meters (mm) wide, while prokaryotes are between 1 and 10 m The term Eukaryotic is derived from the Greek eu, true, and karyon, kernel, referring to the nucleus. Thus Eukaryotic cells are cells with a "true nucleus".

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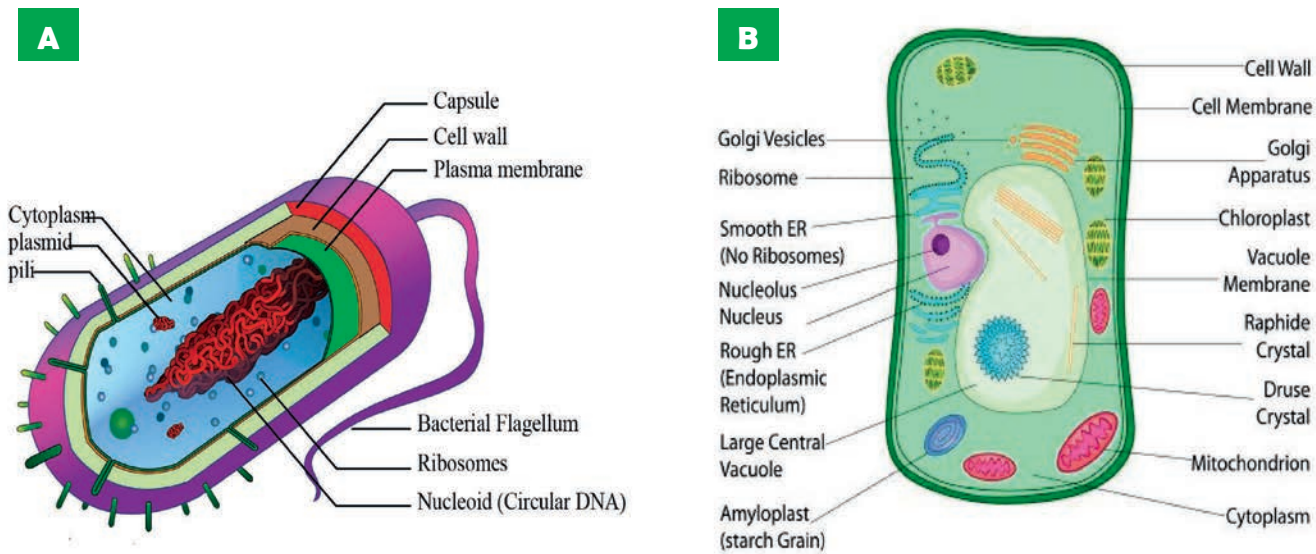


Figure 3.10. A. Structure of a typical prokaryotic cell structure (Example, Bacteria cell) B. Structure of a typical eukaryotic cell (Example, plant cell)

Table 3.1 Comparison of prokaryotic and eukaryotic cells

Structure	Eukaryotic cells	Prokaryotic cells
Organelles	Membrane-bound organelles (for example, nucleus, ER)	No Membrane-bound organelles
Ribosomes	relatively large	relatively small
Chromosomes	DNA arranged in long strands, associated with proteins	DNA present, not associated with proteins, circular plasmids may also be present
Cell wall	always present in plant cells, made of cellulose, never present in animal cells	always present, primarily made of Peptidoglycan
Cilia and flagella	sometimes present	some have flagella, but these have a different structure from those in eukaryotic cells

3.5. Animal and plant cells

Objectives

- At the end of this section, the student will be able to:
- list the structures of plant and animal cells, and
 - describe the differences in structure between plant and animal cells.

Animal and plant cells have certain features in common like cell membrane, nucleus, cytoplasm, mitochondria, and Golgi apparatus. The animal and plant cell structures have also some differences. For example, cell wall, chloroplast and large permanent vacuoles are

found in plant cells while centriole and lysosome are found only in animal cells.

In addition, plant cells are usually larger and are more easily visible under alight microscope than animal cells. The plant vacuole is surrounded by a membrane and contains fluid. The fluid in the vacuole is a solution of pigments, enzymes, sugars and other organic compounds (including some waste products), mineral salts, oxygen and carbon dioxide. Vacuoles help to regulate the osmotic properties of cells (the flow of water inwards and outwards). Chloroplasts are found in the green parts of the plant, mainly in the leaves.

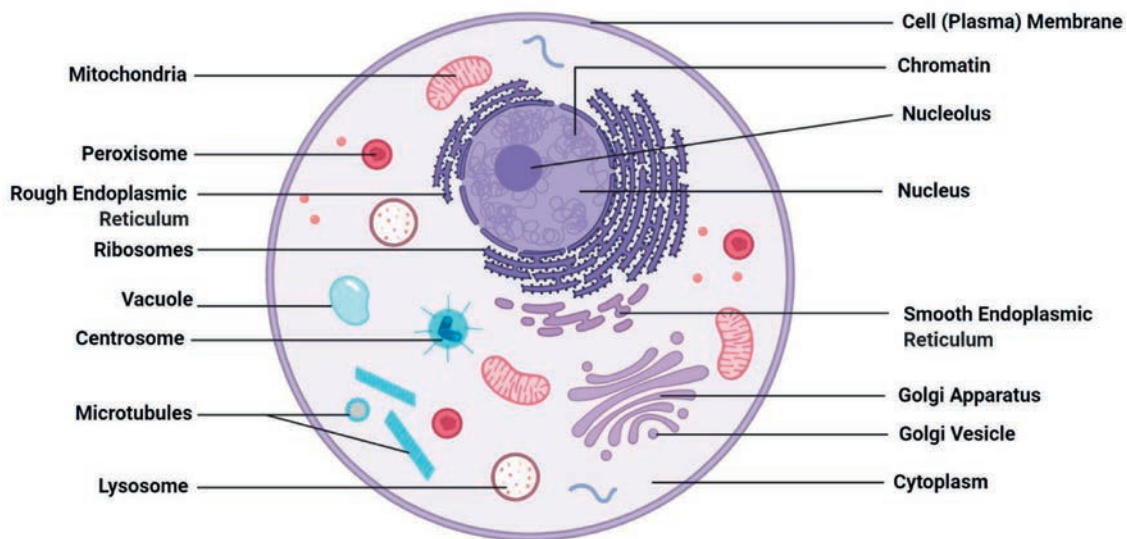
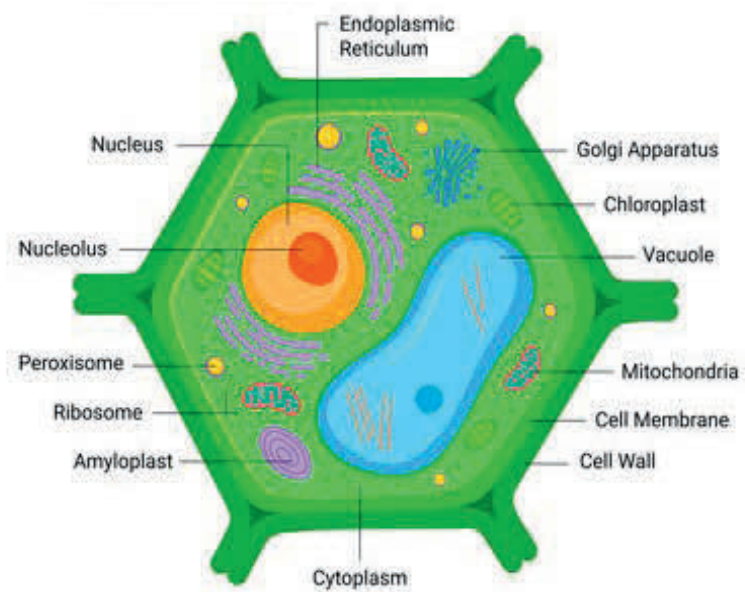
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Fig. 3. 11. A. Animal cell as seen with an electron microscope

B. Plant cells as seen with an electron microscope

Activity 3.11: Self-test

Fill the following table by saying 'present' or 'absent' based on the structure of animal and plant cell.

Features	Plant Cell	Animal Cell
Cell wall		
Cell membrane		
Nucleus		
Chloroplasts		
Vacuole		
Mitochondria		
Rough and smooth endoplasmic reticulum		
Golgi bodies		
Ribosomes		
Lysosome		

3.6 Observing cells under a microscope

Objectives

At the end of this section, the student will be able to:

- draw the cell structures as seen under the light microscope, and
- label the cell wall, cell membrane, cytoplasm and nucleus of the cell.

A microscope is an instrument that can be used to observe small objects, even cells. The image of an object is magnified through at least one lens in the microscope. This lens bends light toward the eye and makes an object appear larger than it actually is.

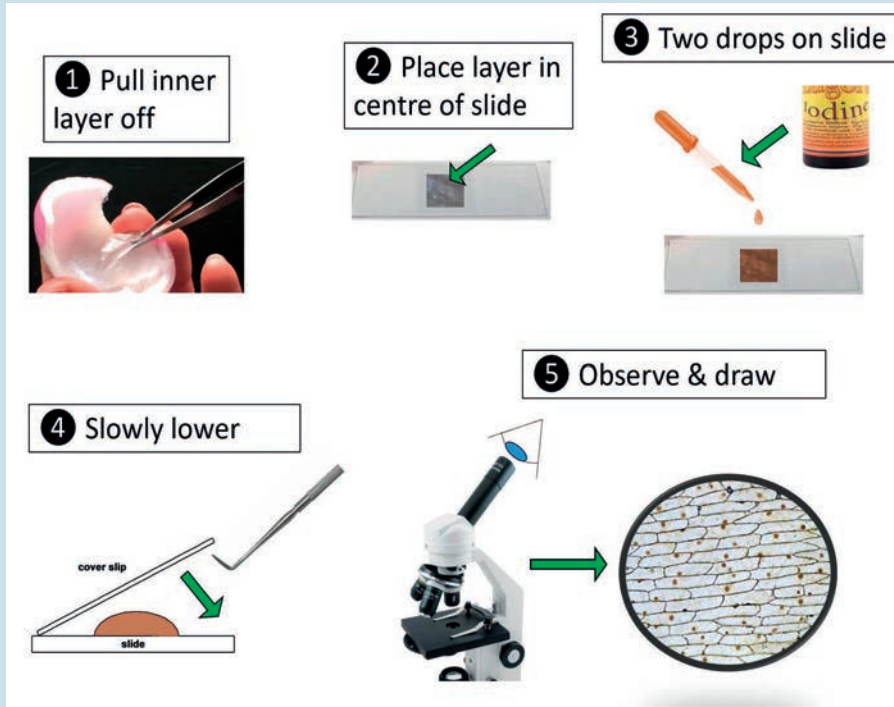
Activity 3.12: Laboratory Activity

In unit one, you have been introduced to the microscope. In this section, you will learn how to observe the structure of plant and animal cells under a light microscope.

Objective: To study the structure of plant cells under a light microscope

For this activity, you will need:

- a microscope
- microscope slides
- cover-slips
- forceps
- a mounting needle
- a pipette
- a lamp
- a piece of onion skin
- iodine solution



Method

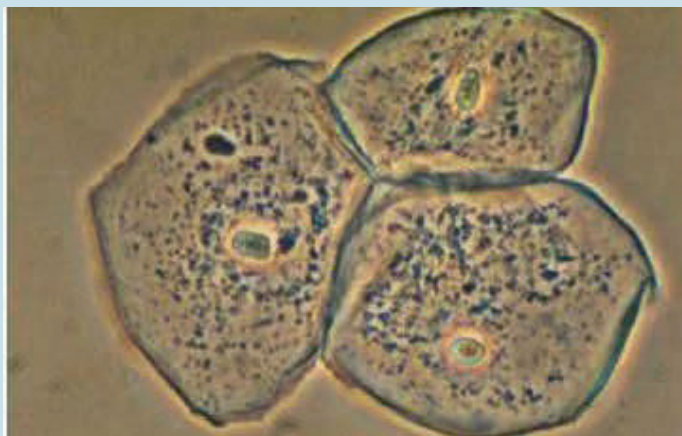
1. Using forceps, peel a piece of epidermal tissue from the inner side of an onion bulb
2. Place the epidermal tissue on a glass microscope slide.
3. Using a scalpel, cut out a 1cm square of tissue (discarding the rest) and arrange it in the center of the slide, Observe the onion cell.
4. Then, add two to three drops of iodine solution for staining. (What do you think the reason for staining? What additional structure can you see in this cell, which could not see in animal cell? Which cellular structure is more visible following staining? Why?)
5. Using forceps, a mounting needle or a wooden splint, support a coverslip with one edge resting near to the onion tissue, at an angle of about 45°.
6. Gently lower the coverslip over the onion tissue, trying to avoid trapping any air bubbles. (Air bubbles will reflect light when viewing under the light microscope, obscuring the features you are trying to observe).
7. Leave the slide for about 5 minutes to allow the iodine stain to react with the specimen.
8. The iodine will stain the cell
9. Place the slide onto the microscope stage, select the lowest power objective lens and focus on the specimen.
10. Increase the magnification using the other objective lenses.
11. Make a large drawing of one cell and label the following parts: cell wall cell membrane, cytoplasm, nucleus. What is the difference between staining and unstaining onion epidermal cell?

Activity 3.13: Laboratory Activity

Objective: To study the structure of animal cells under a light microscope

For this activity, you will need:

- a microscope
- microscope slides
- cover slips
- forceps
- a mounting needle
- a pipette
- a lamp
- cotton bud
- absolute alcohol
- methylene blue dye



Cheek cell

Method

1. Rinse your mouth with water to remove any fragments of food.
2. Take a cotton bud from a freshly opened pack.
3. Rub the cotton bud lightly on the inside of your cheek and gums to collect some cheek cells in saliva.
4. Rub the cotton bud onto the center of a clean microscope slide, to leave a sample of saliva.
5. Repeat if the sample is too small.
6. Then drop the cotton bud into a container of absolute alcohol (why do we use alcohol in this procedure?).
7. Add two to three drops of methylene blue dye. (What do you think the reason for staining by methylene blue? Which cellular structure is more visible following staining? Why?) Using forceps, a mounted needle or wooden splint, support a cover slip with one edge resting near to the cheek cell sample, at an angle of about 45°.
8. Gently lower the cover slip over the tissue, trying to avoid trapping any air bubbles. (Air bubbles will reflect light when viewing under the light microscope, obscuring the features you are trying to observe.)
9. Leave the slide for a few minutes (Why do we stay for a few minutes?)
10. Place the slide onto the microscope stage, select the lowest power objective lens and focus on the specimen.
11. Increase the magnification using the other objective lenses.
12. Make a drawing of one cell and label the following parts: cell membrane, cytoplasm, nucleus.
13. Place your used slide in laboratory disinfectant or autoclaving before washing. Why do we disinfect the used slides?

Note: This technique which involves studying the epithelial cells taken from the inside of the cheek needs some safety rules because of the very small risk of transmitting the AIDS virus. However, appropriate precautions should be applied to treat contaminated items

3.7 The cell and its environment

Objectives

At the end of this section, the student will be able to:

- *define the terms diffusion and osmosis,*
- *describe the importance of diffusion and osmosis for the transport of gases and solutes, and*
- *describe the importance of active transport.*

The Cell membrane

As previously mentioned, the cell membrane is a selectively (semi) permeable structure that controls what enters and leaves cells.

Activity 3.14 Think-pair share

1. If the cell membrane is to lose its selectivity, what do you think will happen to the cell? Discuss in pairs and share your answers with your classmates.
2. List some of the substances which leave and enter the cell

Cells need food materials to get energy or to build up its cell structures. Cells also need salts and water, which play a part in chemical reactions in the cell. Cells also need to remove unnecessary substances such as carbon dioxide.

The structure of a cell known as cell membrane is semi-permeable, it controls the entering of necessary substances and the removal of unnecessary ones. Materials can pass through the cell membrane either passively by diffusion and osmosis or actively by active transport.

3.7.1 Passive transport

Passive transport is the movement of substances down a concentration gradient. This movement is from an area of high concentration to an area of lower concentration without the need for input of energy.

Attention

You can think of passive and active transport like rolling a ball on a hill. Imagine that the hill is made of molecules; the lower part of the hill has few molecules, while the upper part of the hill has lots of stacked molecules. If you want to roll a ball downhill (or let molecules flow from where they're piled up to where they're not), you don't have to use any energy. If you want to roll the ball uphill (or push molecules from where they're in low concentration to where they're in high concentration), you have to put in some energy.

Attention

Thinking of the plasma membrane as your school guard, who is always standing at the entrance of the school gate and control who and what enters and leaves the school, which is a good way of remembering the cell membrane's function.

Key Terms

Diffusion: is the net movement of molecules or ions from a region of high concentration to a region of low concentration

Osmosis: is the diffusion of water molecules from a region of their higher concentration (dilute solution) to a region of their lower concentration (concentrated solution).

Active transport: is the movement of ions or molecules across the cell membrane, against a concentration gradient, using energy.

Diffusion

This is the movement of molecules from a region of high concentration to a region of low concentration down the concentration gradient. Diffusion is a passive transport of molecules; the molecules need only spread themselves, or diffuse, across the membrane. The diffusion of gases and solutes is important for any activity of the cell. For example, oxygen is important for respiration and get into the cell by diffusion.

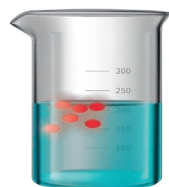
Activity 3.15: Think– Pair –Share

Discuss in pair what will happen in the following situations, Discuss with your friend and share your idea with the other class-mates.

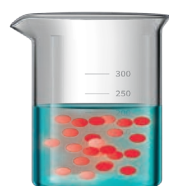
1. When some odor, like incense stick is placed in one part of our home
2. When tea bag is dipping in a glass of hot water



(a) If a drop of colored ink is dropped into a beaker of water



(b) its molecules dissolve



(c) and diffuse



(d) Eventually, diffusion results in an even distribution of ink molecules throughout the water.

Figure 3.12. Diffusion of molecules.

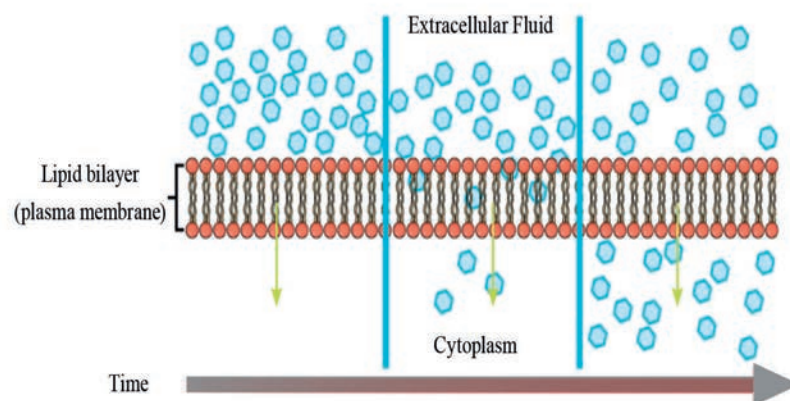


Figure . 3.13 Diffusion through a semi-permeable membrane of cell.

Factors influencing the rate of diffusion:

Concentration gradient: the higher the difference in the concentration of a substance on either side of a membrane, the faster it will tend to diffuse from higher concentration to lower.

Temperature: increase in temperature causes an increase in the kinetic energy that molecules and ions possess enabling them to move faster.

Mass of the molecule: heavier molecules move more slowly; therefore, they diffuse more slowly while lighter molecules diffuse faster.

Distance travelled: the farther the distance that a substance must travel, the slower the diffusion rate.

Surface area: The greater the surface area, the faster the total diffusion is. Cells that are involved in rapid absorption, such as those in the kidney or the intestine, often have their 'free' surface membrane formed into hundreds of tiny projections called microvilli which increase the absorbing surface.

Cells increase their surface area
by in-foldings of the membrane
like these microvilli

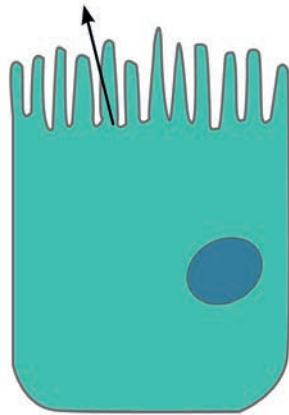


Figure. 3. 14 Microvilli on the surface of small intestine

Activity 3.16: Group Work

Select one student for this activity, follow the instruction below and discuss your findings with your classmates.

Objective: To study the diffusion of perfume odour from high concentration towards the lower concentration.

For this activity, you will need: Bottle of perfume

Method

1. Ask your friend to spray some perfume on the front side of your class.
2. Another student should counting number as the smell of the perfume diffuse through the classroom
3. Each student should hand up when he/she smell the odor of the perfume.
4. You'll see a wave of hands moving from the front to the back and sides of the class as the molecules spread out by diffusion.
5. Discuss, how different students smell the odor of the perfume at different times, how long it takes to reach the student in the last row?
6. Repeat the activity by increasing the concentration of the perfume and compare the time with the previous one
7. Do you get a different result when you increase the concentration of the perfume?

Unit 3: Cells

Activity 3.17: Think– Pair –Share

Discuss with your friend and share your idea with the other classmates why this happen in the following situations

1. Feeling thirsty after having salty food.
2. Swelling of resins and other seeds when they are soaked in water
3. If you put a potato into pure water, it swells up over time.

Key Terms**Hypertonic solution:**

a solution with a high concentration of solute and a low concentration of water.

Hypotonic solution:

a solution with a low concentration of solute and high concentration of water.

Isotonic solution: when the cell and its surroundings have equal solute and water

Attention**Application of osmosis in day to day life**

People have used osmosis to preserve food like meat and fish through salt preservation.

Osmosis

Osmosis is a special case of diffusion. Like other substances water can also move passively through cell membrane. It moves from a region of high concentration of water molecules (lower solute concentration) to a region of lower concentration of water molecules (higher solute concentration).

Activity 3.18: Think– Pair –Share

Think about osmosis in terms of the solutes. Why do you think water moves toward the area with the greatest concentration of solutes?

Discuss with your friend and share your idea with the other classmates.

The movement of water is affected by the concentration of substances (solutes) dissolved in the water. Basically, water moves from areas where there is more water to areas where less water is.

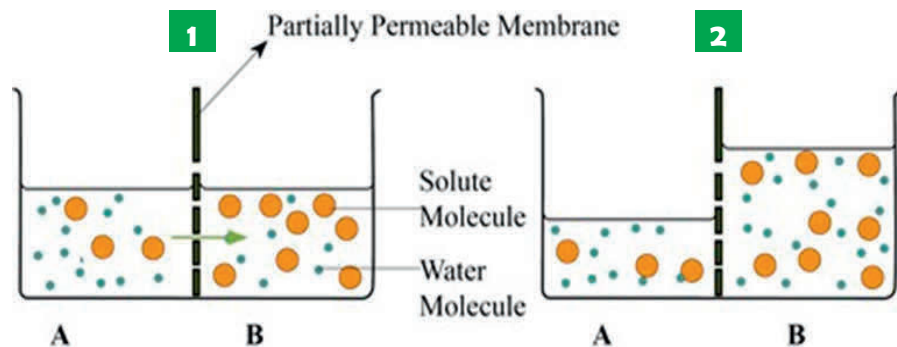


Figure. 3.15 Osmosis

1. Before osmosis: two solutions are separated partially permeable membrane. B has a higher solute concentration than A. The soluble molecules are too large to pass through the pores in the membrane but the water molecules are small enough.
2. As the arrows in diagram A indicate, more water molecules moved from A to B than from B to A, so the net movement has been from A to B, raising the level of the solution in B and lowering in A.

The movement of water into and out of the cell depends on the type of solution found in the cell surrounding (Environment found outside the cell).

When a cell is in a hypertonic solution, water moves out of the cell, causing the cell to shrink or plasmolysed. In an isotonic solution, water diffuses into and out of the cell at the same rate, with no change in cell size. In a hypotonic solution, water moves into the cell and pressure are applied to the plasma membrane causing turgidity of the cell. The strong cell wall of plant cells can resist this pressure to keep the cell from rupturing. This is not the case with animal cells as animal cells lack cell wall.

Activity 3.19: Group work

Discuss in groups how our society uses the principles of osmosis for the preservation of meat in the rural area where the refrigerator is not available.

Osmosis in Plant cells

Osmosis in plants is very important to bring strength and rigidity to the plant cell. If a plant cell is surrounded by water or a solution more dilute than its contents (hypotonic solution), water will pass into the vacuole of the plant. The vacuole will expand and press outwards on the cytoplasm and cell wall. Then, the rigid cell wall expands and exert the pressure back, preventing the cell from taking in too much water and bursting. The plant becomes very turgid, due to the pressure of water pressing outwards on the cell wall.

Turgor pressure is necessary for plants to retain their upright posture and the extended state of their leaves. In contrast, if a plant cell is surrounded by water or a solution less dilute than its contents (hypertonic solution), plant cells lose water and there will no longer be any water pressure pressing outwards against the cell walls, the plasma membrane shrinks pulls away from the cell and the cell get plasmolysed. The plasmolysed cell wilts and eventually dies.

In this figure 3.16 A. shows that if the water concentration of the solution surrounding the plant cell is too high, the cell swells and turgid. B. If it is the concentration water in both is the same C. If the solution surrounding the cell is too high, the plant cell plasmolysed.

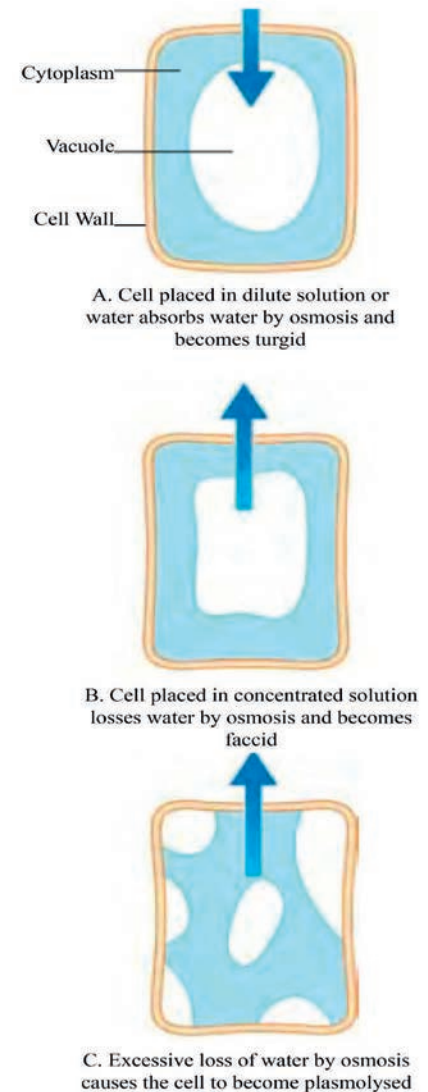


Fig. 3.16. Osmosis in plant

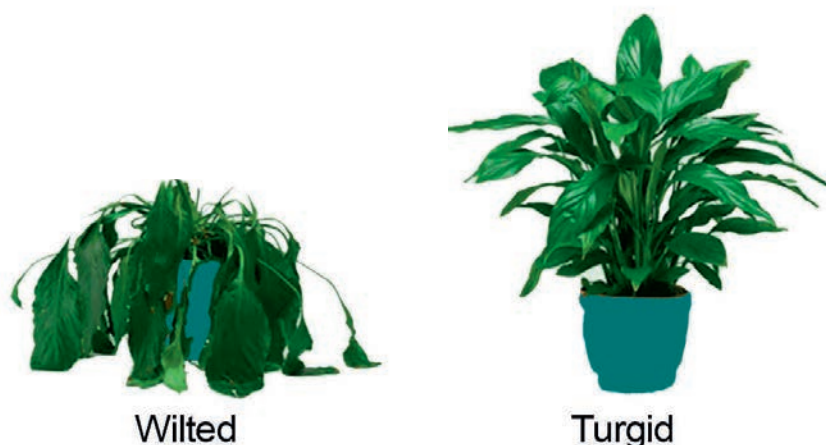


Figure 3.17. The effect of osmosis in plant leaf

Without adequate water, the plant on the left has lost turgor pressure, visible in its wilting. Watering the plant (right) will restore the turgor pressure.

Osmosis in Animal Cells

The excessive uptake or loss of water by animal cells causes damages to cells. The survival of an animal cell, thus, depends on its ability to balance water uptake and loss.

In the case of animals particularly vertebrates, the concentration of water in the blood is monitored by the brain and adjusted by the kidneys. The regulation of water flow is by keeping the blood concentration within narrow limits, the concentration of the tissue fluid remains more or less constant.

The animal cell should be surrounded by fluid or blood plasma which has the same concentration of water as the cell contents; there is no net flow of water into or out of the cells. If not balanced, osmosis can cause a serious problem in the animal cell. For Example, if the surrounding solution has a high concentration of water (hypotonic) than the cells, water will move into the cells by osmosis.

Water entering the cell will make the cell swell up. Animal cells have no cell wall and the membrane has little strength, water would continue to enter and the cells will eventually burst. This process is called hemolysis of the cell which can occur in red blood cells. Conversely, if the surrounding solution has a low concentration of water (hypertonic) than the cells, water will pass out of the cell by osmosis and the cell will shrink.

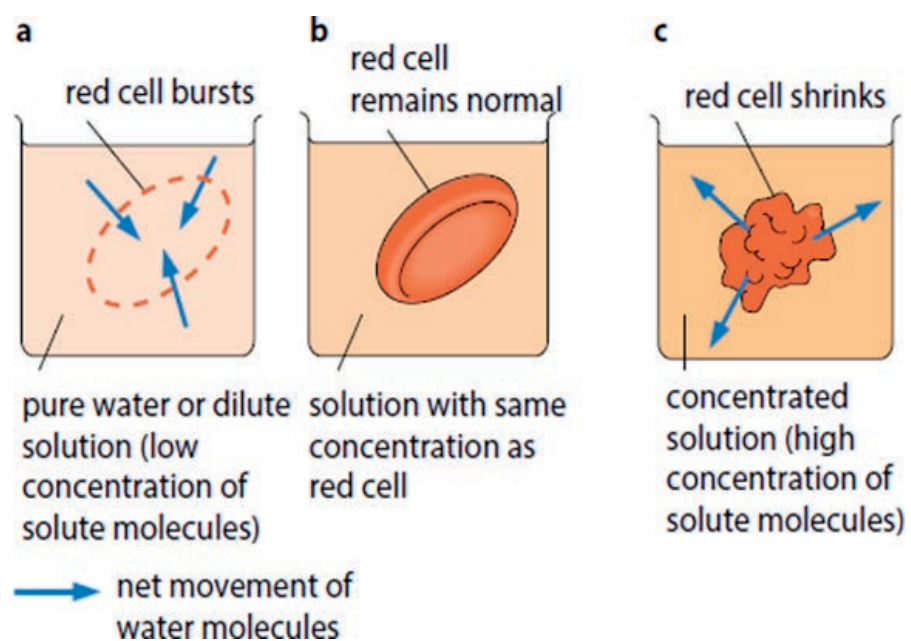
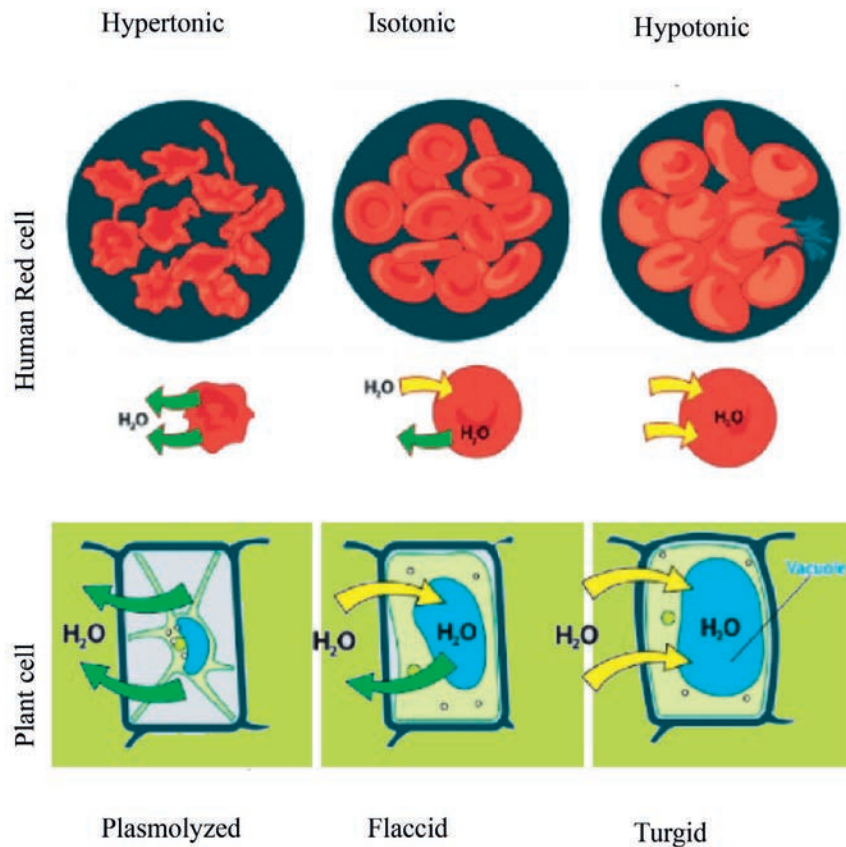


Fig. 3.18. (A) Shows that if the water concentration of the solution surrounding the red blood cell is too high, the water moves to the cell and the cell swells and bursts. (B) If the water concentration in red blood cell and the surrounding solution is the same. (C) If the water concentration of the solution surrounding the red blood cell is low, the water moves from the cell to the surrounding solution and the cell shrinks

**Activity 3.20: Group work**

1. Make a group of five students and discuss the conditions that derive us to drink water after a heavy physical exercise?
2. A farmer cultivates a crop plant and he applies too much chemical fertilizers to the soil, does this affect the crop plant? What could happen the crop plant? Why?

Fig.3.19 Osmosis in plants and animal cell

Activity 3.21: Homework /Individual assignment

Using information from the internet or book, discuss how animals living in freshwater regulate osmosis and how they protect their body from bursting?

Osmosis in Ameba

Unicellular protozoans such as amoeba have contractile vacuoles. Contractile vacuole are membranous structures which controls the intracellular water balance by accumulating and expelling excess water out of the cell, allowing cells to survive under hypotonic stress as in pond water participate in osmoregulation of the cell and prevent the bursting of cells.

The unicellular protozoans which dwell in freshwater consume excess water, with the help of contractile vacuoles they drain excess water out. It also helps them to protect the cell in the absence of cell wall. Without these vacuoles, they might take in too much water, swell and burst out.

Activity 3.22: Laboratory Activity

Osmosis in plant cells

Objective: To Investigating osmosis in plant cells

For this activity, you will need:

- Potato,
- salt solution,
- Balance

1. Observing osmosis in potatoes:

- prepare equal sized potato cubes and measure the weight
- make a salt solution
- bring another bottle and put pure water in it
- put the potato cubes in pure water and salt solution. Speculate what will happen?
- After 1hour observe their difference in size, weight, and conclude.

2. Observing osmosis in onion cell

Objective : To study plasmolysis using onion cell.

Epidermal strips are useful material for observing plasmolysis. Colored sap makes the observation easier. Suitable sources are the inner surfaces of the fleshy storage and leaves of red onion bulbs.

- Place the strips of the epidermis in a salt or sugar solution.
- Place small pieces of the strips on glass slides.
- Mount in the relevant solution.
- Observe with a microscope.
- What you observe about the shape of onion cell?
- Draw the picture of the cell that you observe under microscope
- Plasmolysis may take several minutes, then discuss in groups about your result.

Activity 3.23: Laboratory Activity

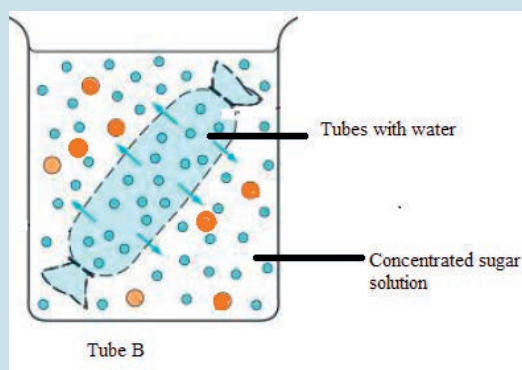
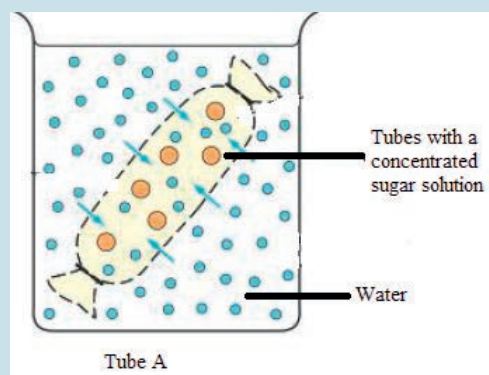
Investigating osmosis using model cell. Dialysis tubing is partially permeable. We can use dialysis tubing to represent the cell membrane and the solution inside it to represent the cytoplasm.

Objectives: Objective: To Investigating osmosis in animal model cells

For this activity, you will need:

- Dialysis tube,
- sugar solution,
- beaker.

1. Cut two pieces of dialysis tubing, each 15cm long. Tie one end.
2. Fill one dialysis tube with a concentrated sugar solution (tube A). Fill the other model cell with water (tube B).
3. Put 'cell A' into a beaker of water and put 'cell B' into a beaker containing a concentrated solution of sugar.
4. After 20 to 30 minutes take out the two tubes. What happen to the two tubes? Is there any difference in volume between tube A and tube B?



Attention

Active transport allows cells to maintain internal concentrations of small solutes that differ from environmental concentrations. For example, compared with its surroundings, an animal nerve cell has a much higher concentration of potassium ions and a much lower concentration of sodium ions. The plasma membrane helps maintain these differences by pumping sodium out of the cell and potassium into the cell. This particular case of active transport (called the sodium-potassium pump) is vital to the nervous system of most animals.

3.7.2 Active transport

Active transport is the movement of molecules from areas where they're less concentrated to areas where they're more concentrated using energy. Active transport enables cells to use their stored energy to concentrate molecules inside or outside of the cell.

Active transport enables cells to use their stored energy to concentrate molecules inside or outside of the cell. Active Transport is the movement of particles through a cell membrane from a region of lower concentration to a region of higher concentration against a concentration gradient, using the energy released during respiration. Cells sometimes take in substances even though there is already a higher concentration inside than outside. The substance is absorbed against the concentration gradient. Less commonly, cells sometimes pump substances out, even though there is already a larger concentration outside. Some of the examples are: ion uptake by root hairs and uptake of glucose by epithelial cells of villi.

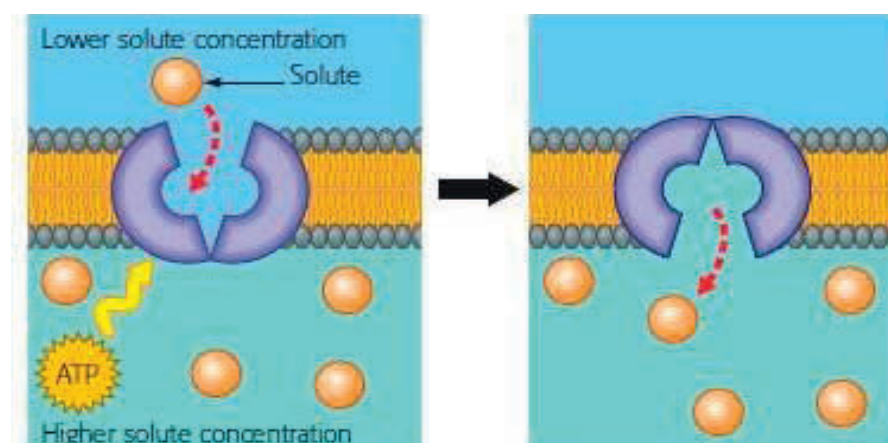


Fig 3.20 Active transport

3.8. Levels of Biological Organization**Objectives**

At the end of this section, the student will be able to:

- *define the terms tissue, organ and organ system, and*
- *list some examples of human systems.*

Activity 3.24: Think – Pair – Share

What is the smallest level of biological organization that can display all the characteristics of life? Discuss with a partner and share your answers with your classmates.

Life forms a hierarchy of organization from atoms to complex multicellular organisms. Atoms are joined together to form molecules, which are assembled into more complex structures such as organelles. These in turn form subsystems that provide different functions. Cells can be organized into tissues, then into organs and organ systems such as the nervous system. This organization then extends beyond individual organisms to populations, communities, ecosystems, and finally the biosphere.

Activity 3.25: Think – Pair – Share

Discuss with a partner and share your answers with your classmates.

1. How do these different cells and relate to each other? Where do body systems, like the digestive system and nervous system fit into levels of organization?
2. Suppose one organ in a digestive system failed to work properly. How would the digestive system be affected? Would it still be able to function? Explain your reasoning.
3. What might happen if a plant organ, such as a leaf or stem, became damaged? How would the organism be affected?

The organization of the biological world is hierarchical—that is, each level builds on the level below it:

The cellular level. At the cellular level, **atoms**, the fundamental elements of matter, are joined by s chemical bond into clusters called **molecules**. Complex biological molecules are assembled into tiny structures called **organelles** within membrane-bounded units we call **cells**.

Most cells, when they have finished dividing and growing, become specialized. When cells are specialized, they do one particular job, they develop a distinct shape and special kinds of chemical changes take place in their cytoplasm

The organism level. Cells in complex multicellular organisms exhibit three levels of an organization. The most basic level is that of **tissues** (for example nerve tissue), which are groups of similar cells that act as a functional unit. The cells of each type have a similar structure and function so that the tissue itself can be said to have a particular function; for example, muscles contract to cause movement, xylem carries water in plants. The tissues of the leaf include the epidermis, palisade tissue, spongy tissue, xylem and phloem.

Tissues, in turn, are grouped into **organs** body structures composed of several different tissues that act as structural and functional units. For example, your brain is an organ composed of nerve cells. Organs consist of several tissues grouped to make a structure with a special function.

For example, the stomach is an organ that contains tissues made from epithelial cells, gland cells and muscle cells. These cells are supplied with food and oxygen brought by blood vessels. The heart, lungs,

Key Terms

Tissues: group of cells that have the same function.

Organs: group of tissues that work together to do a particular function.

Systems: different organs which are working together

Unit 3: Cells

intestines, brain and eyes are further examples of organs in animals. In flowering plants, the root, stem and leaves are the organs.

At the third level of organization, organs are grouped into **systems**. The digestive system, for example, consists of digestive organs like the stomach and intestine. The different systems collectively form an organism. An organism is formed by the organs and systems working together to produce an independent plant or animal.

Activity 3.26: Group work

Give examples of some systems in the human body and discuss The organs from which they are composed of

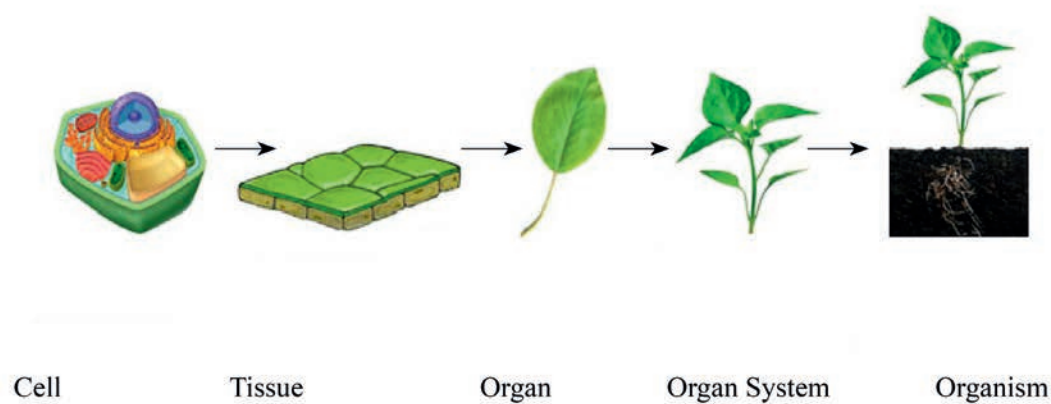


Figure. 3.21 Level of organization in plants

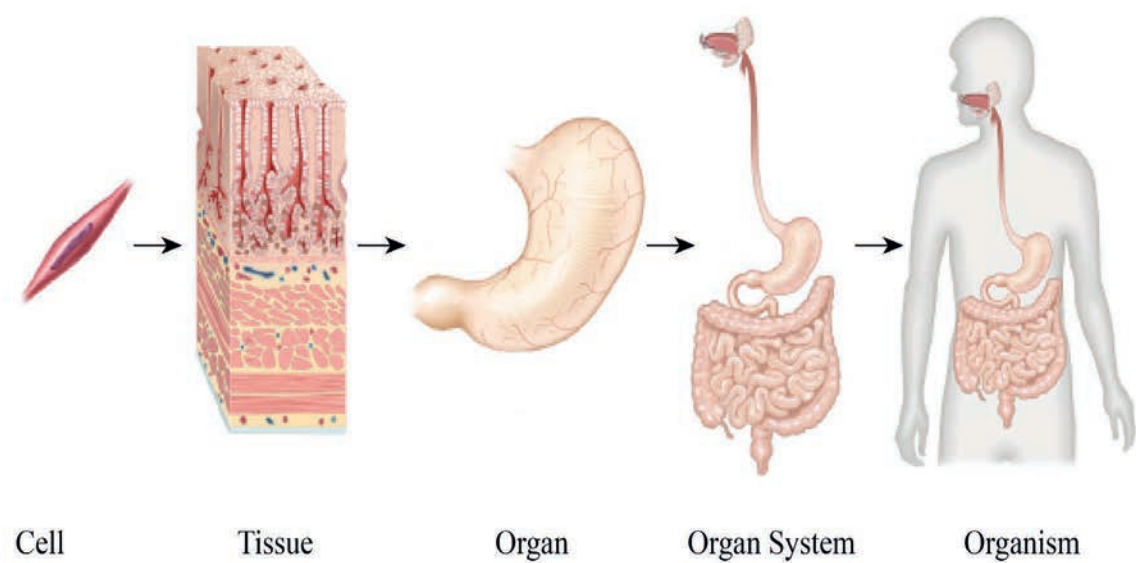


Figure. 3.22 Level of organization in Animals

Unit review

In this unit, you have learnt about:

- Cell theory is the unifying foundation of cell biology. All organisms are composed of one or more cells. Cells arise only by the division of preexisting cells.
- All cells share simple structural features.
- All eukaryotic cells have a centrally located nucleus, a semifluid cytoplasm, and an enclosing plasma membrane.
- In eukaryotic cells, membranes partition into several functional compartments called organelles. The largest organelle is usually the nucleus. Other organelles are located in the cytoplasm, the region outside the nucleus and within the plasma membrane.
- Nucleus is the structure of the cell which control all activities of the cell. Within the nucleus, DNA and proteins make up chromatin fibers; each very long fiber is a single chromosome. The nucleus also contains the nucleolus, which produces components of ribosomes.
- Endoplasmic Reticulum is the structure consists of membrane-enclosed tubes and sacs within the cytoplasm. Rough ER, named because of the ribosomes attached to its surface, makes membrane and secretory proteins. The functions of smooth ER include lipid synthesis and detoxification.
- The Golgi apparatus refines certain ER products and packages them in transport vesicles targeted for other organelles or export from the cell.
- Lysosomes are sacs containing digestive enzymes, aid digestion and recycling within the cell.
- Vacuoles include the contractile vacuoles that expel water from certain freshwater protists and a large multifunctional central vacuoles of plant cells. Microtubules are an important component of the cytoskeleton, an organelle that gives support to and maintains the shape of, cells.
- Cilia and eukaryotic flagella are appendages that aid in movement.
- Animal and plant cells have a cell membrane, nucleus, cytoplasm and organelles like mitochondria (singular: mitochondrion), Golgi apparatus.
- The difference between animal and plant cells is the presence of centriole, chloroplast, cell walls is large permanent vacuoles only in plant cells.
- Passive transport is the movement of substances down a concentration gradient. This movement is from an area of high concentration to an area of lower concentration without the need for energy to be used. Whereas active transport is the movement of molecules from areas where they're less concentrated to areas where they're more concentrated using energy. Life forms a hierarchy of organization from atoms to complex multicellular organisms. Atoms are joined together to form molecules, which are assembled into more complex structures such as organelles. These in turn form subsystems that provide different functions. Cells can be organized into tissues, then into organs and organ systems such as the nervous system. This organization then extends beyond individual organisms to populations, communities, ecosystems, and finally the biosphere.

Review Questions

i. Match the following organelles under column 'B' with their functions under column 'A'

A	B
1. Protein export	a. Mitochondria
2. Control cellular activity	b. Golgi apparatus
3. Digestion	c. Nucleus
4. Powerhouse of the cell	d. Lysosome
5. Photosynthesis	e. Chloroplast

ii. Choose the best answer for the following questions

- Which one of the following structures is found in animal cells, but not in plant cells?
A. cell surface membrane B. centriole C. chloroplast D. Golgi body
- When the cell or the surrounding environment contains a relatively high concentration of water, we call it:
A. isotonic B. hypertonic C. hypotonic D. Osmotic
- Which structure is found in plant cells but not in animal cells?
A. Cell membrane B. Mitochondrion C. Cell wall D. Nucleus
- In which direction does the transport of water occur when a cell containing 1% sodium chloride is placed in seawater that contains 10 % sodium chloride?
A. From the cell to seawater B. From seawater to cell
- If you look into a microscope and view an unknown cell. What structure might you see that would tell you that the cell is eukaryotic?
A. DNA B. nucleoid region C. a plasma membrane D. membrane-enclosed structures called organelles.

iii. Fill in the blank space with appropriate words

- A structure that supports and strengthen plant cell is _____.
- A structure that breaks down a captured bacterium and damaged organelle in the cell is_____.
- The sac-like structure used to transport protein is_____.
- A structure made up of a group of tissues, working together to perform a specific Function is_____.
- When the osmotic pressure is the same on both sides of a semipermeable membrane, each solution is said to be _____.

iv. Answer the following questions.

- What would happen to a person's red blood cells if the person is given an injection of pure water (0% saline)?
- Name three structures in plant cells that animal cells lack.

- 3. Name two structures that may be found in animal cells but not in plant cells.
- 4. List the structural features that prokaryotic and eukaryotic cells have in common. Briefly explain why each of the structures you have listed is essential.
- 5. Assume that you are observing animal cell under the microscope and the animal cell looks shrives, what type of environment or solution is found around this animal cell.

1. Fill the following table space by saying YES or No

Cell structure s	cheek cell (animal)	Onion cell (plant)
Nucleus		
cell wall		
chloroplasts		
large vacuole		
Cytoplasm		

2. Fill the blank spaces of the table with appropriate level organization of some organ system

Cell	Tissue	organ	system
Red blood cell	cardiac muscle	Heart	
Egg, sperm	Epithelial tissue of male and female reproductive organ		Reproductive system
Mucous cell	Epithelial digestive tissue		Digestive system
Xylem or phloem cells	Xylem /phloem	root /stem	
Nerve cell	Nerve tissue		Nervous system

3. Fill in Fill the blank spaces of the table by saying YES or NO

	Diffusion	Osmosis	Active transport
It is the movement of substances like O ₂			
It is movement of water			
Movement of ions like K ⁺			
Energy is required			